Faculty of Engineering

Course of Study: Industrial Artificial Intelligence

Industrial AI: Mini Project 2

Predictive Maintenance for Ball Bearings

CNN-Based Remaining Useful Life Prediction on FEMTO Data

A CRISP-DM Approach

Abstract

Ball bearing failures are the primary cause of unplanned downtime on production lines with heavily loaded electric motors, resulting in significant production delays, emergency repair costs, and missed delivery commitments. This project addresses that challenge by developing a predictive maintenance system for bearing Remaining Useful Life (RUL) prediction using the FEMTO/PRONOSTIA dataset from the IEEE PHM 2012 Challenge. The system was developed following the CRISP-DM methodology and combines a one-dimensional Convolutional Neural Network (CNN) for vibration pattern extraction with a Support Vector Regression (SVR) model for RUL prediction, and a Random Forest classifier for health stage classification. Five statistical features were extracted from bandpass-filtered vibration windows per channel, yielding a ten-feature input matrix. A Health Indicator (HI) was constructed via Principal Component Analysis (PCA) fusion of RMS and kurtosis trends to track degradation from 0 to 1 across bearing lifetime. Model evaluation used Leave-One-Bearing-Out (LOBO) cross-validation, the standard approach for this dataset. The CNN & SVR pipeline achieved a mean R^2 of 0.399 and mean MAE of 0.35 hours excluding the lifetime-outlier bearing B1_1, which is consistent with published benchmarks on this dataset. The Random Forest classifier achieved a mean F1 macro score of 0.363, representing a ΔF1 improvement of +0.234 over the majority-class baseline. Results are presented through a maintenance dashboard that answers three operational questions: how many hours until replacement is needed, what health stage the bearing is currently in, and how confident the estimate is.

Introduction

Rolling element bearings are fundamental components in electric motors used across manufacturing environments. Their failure is responsible for a disproportionate share of unplanned downtime in industrial settings. Unlike scheduled preventive maintenance, which replaces components at fixed intervals regardless of actual wear state, predictive maintenance aims to identify the onset of degradation and estimate the time remaining before a failure occurs, enabling interventions to be scheduled at precisely the right moment.

This project addresses the following operational question: given continuous vibration monitoring data from a running bearing, can a model accurately predict how many hours of useful life remain? The work is framed as both a regression problem (predicting RUL in hours) and a classification problem (assigning a health stage label), with an additional requirement to quantify prediction uncertainty in a form useful to a production engineer rather than a data scientist.

The FEMTO/PRONOSTIA dataset (Nectoux et al., 2012), provided by the FEMTO-ST Institute and used in the IEEE PHM 2012 Challenge, was selected as the experimental basis. The dataset contains run-to-failure vibration recordings from ball bearings under three operating conditions, making it a standard benchmark for RUL prediction research. The data-driven approach taken in this work follows Cheng et al. (2018), who proposed a CNN-based degradation indicator combined with SVR regression and demonstrated its effectiveness on exactly this dataset.

The remainder of this report is structured according to the CRISP-DM framework: Section 2 describes the dataset and data understanding; Section 3 covers data preparation and feature engineering; Section 4 presents the modelling approach; Section 5 reports evaluation results; Section 6 describes the maintenance dashboard deployment; and Section 7 discusses conclusions, limitations, and future work.

Data Understanding

Dataset Description

The FEMTO/PRONOSTIA dataset consists of accelerated run-to-failure bearing experiments conducted on the PRONOSTIA test platform at the FEMTO-ST Institute in Besançon, France. The dataset was released as part of the IEEE PHM 2012 Prognostics Challenge and has since become a widely-used benchmark for bearing RUL prediction.

Bearings were tested under three operating conditions defined by rotational speed and radial load: Condition 1 (1800 rpm, 4000 N), Condition 2 (1650 rpm, 4200 N), and Condition 3 (1500 rpm, 5000 N). The learning set, used for model training, contains two run-to-failure bearings per condition, yielding six training bearings. Due to the split structure of the rul-datasets library used to load the data (Krokotsch, 2023), five bearings were available in the development split: B1_1, B1_2, B2_1, B2_2, and B3_1.

Each bearing was instrumented with two piezoelectric accelerometers measuring horizontal (H) and vertical (V) vibration acceleration. The data acquisition protocol records a 0.1-second vibration burst every 10 seconds throughout the bearing's entire operational life. Each burst is sampled at 25.6 kHz, yielding 2,560 samples per window. Temperature signals were also recorded for some bearings but were excluded from this analysis due to inconsistent availability across conditions, consistent with standard practice in the literature.

Exploratory Data Analysis

Figure 1 shows the raw vibration signal for the first and final window of each training bearing. Healthy windows exhibit low-amplitude, symmetric oscillations typically within ±2–3 g. Near-failure windows show markedly higher amplitude and impulse-like spikes, with B1_1 reaching peak amplitudes exceeding ±50 g at failure. This visual contrast confirms that fault-related information is present in the raw signal and motivates the feature extraction approach described in Section 3.

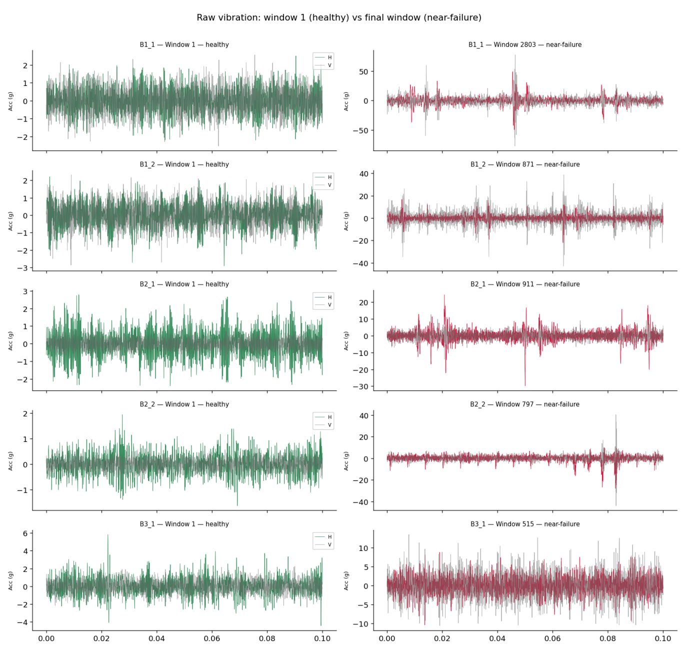

Figure 1. Raw vibration signal comparison: first window versus final window per bearing.

Figure 2 shows the RMS degradation trend over the full lifetime of each bearing using the horizontal channel. The RMS signal is broadly flat during the healthy phase and increases sharply in the final stages before failure. The dashed vertical line marks the end of the first 10% of each bearing's life, which is used as the healthy baseline for Health Indicator construction. The figure also reveals the key data challenge: bearing lifetimes range from 1.43 hours (B3_1) to 7.78 hours (B1_1), a factor of approximately 5.4. This scale mismatch is a primary reason why raw RUL values cannot be used as a regression target across bearings without normalisation.

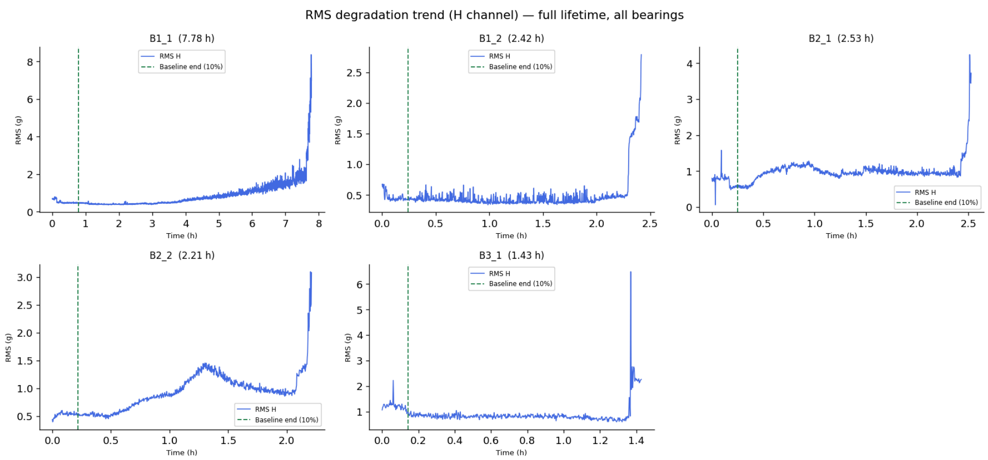

Figure 2. RMS degradation trends over bearing lifetime. Dashed line marks the end of the healthy baseline window.

Table 1 summarises the key dataset statistics.

Bearing	Windows	Lifetime (h)	Condition	Speed / Load
B1_1	2803	7.78	1	1800 rpm / 4000 N
B1_2	871	2.42	1	1800 rpm / 4000 N
B2_1	911	2.53	2	1650 rpm / 4200 N
B2_2	797	2.21	2	1650 rpm / 4200 N
B3_1	515	1.43	3	1500 rpm / 5000 N

Table 1. FEMTO/PRONOSTIA dataset summary for the development split.

Data Preparation and Feature Engineering

Signal Filtering

Each vibration window was processed with a fourth-order Butterworth bandpass filter applied in the frequency range 2–10 kHz. This range was selected to isolate bearing fault frequencies while excluding shaft harmonics below 2 kHz and high-frequency electromagnetic noise above 10 kHz, consistent with the approach of Sutrisno et al. (2012) and Cheng et al. (2018). The filter was applied in zero-phase mode using the filtfilt function to avoid time-shift distortion.

Statistical Feature Extraction

Five time-domain statistical features were computed per channel from the filtered signal, yielding ten features per window across horizontal and vertical channels. The features and their physical interpretation are as follows:

- **Root Mean Square (RMS):** captures the overall energy of the vibration signal. It is the primary degradation indicator identified in the literature (Sutrisno et al., 2012) and consistently ranked as the most important feature in the Random Forest importance analysis (23.2% for the horizontal channel).
- **Kurtosis:** measures the peakedness of the signal distribution relative to a normal distribution. A healthy bearing produces relatively symmetric, low-kurtosis vibration. Early-stage faults generate periodic impulse events that increase kurtosis significantly before any change in RMS is detectable, making it the most sensitive feature for early fault onset detection.
- **Crest Factor:** is the ratio of peak amplitude to RMS. Unlike RMS, which increases monotonically with degradation, crest factor decreases in later fault stages as the overall energy rises. The opposing trends of RMS and crest factor together provide confirmation of advanced degradation.
- **Skewness:** quantifies distributional asymmetry. A healthy bearing vibration signal is approximately symmetric around zero (skewness ≈ 0). The development of directional fault forces introduces asymmetry, causing skewness to shift from zero. This feature provides information independent of kurtosis, which measures peakedness rather than direction.
- **Impulse Factor:** is the ratio of peak amplitude to mean absolute value. It is more sensitive to individual sharp spikes than crest factor and has been identified as a top-three feature in SHAP-based interpretability studies on bearing fault diagnosis.

All features were computed as lambda functions applied to the filtered 1D signal arrays, enabling vectorised computation. The complete feature matrix X had shape (5897, 10) across all five bearings.

Health Indicator Construction

A key challenge in cross-bearing RUL prediction is that different bearings under different operating conditions produce different absolute vibration levels. A raw RMS value of 1.5 g may indicate early degradation for B1_1 but advanced wear for B3_1. To resolve this, a normalised Health Indicator (HI) was constructed to map each bearing's degradation trajectory to the interval [0, 1] regardless of operating condition.

The HI was constructed by fusing the RMS-H and Kurtosis-H feature trends via Principal Component Analysis (PCA). Both feature series were first smoothed using a 15-window rolling mean to reduce window-to-window noise. The smoothed series were then standardised and stacked as a matrix of shape (n_windows, 2). PCA with a single component was applied; the first principal component (PC1) captures the shared degradation direction across the two features. PC1 was then normalised to [0, 1] and additionally smoothed with a five-window rolling mean.

The explained variance ratio for PC1 ranged from 55.5% (B2_1) to 90.3% (B2_2), indicating that RMS and kurtosis are substantially co-aligned in their degradation behaviour. The HI was used for visualisation in the maintenance dashboard and is not used directly as a model input or class label source.

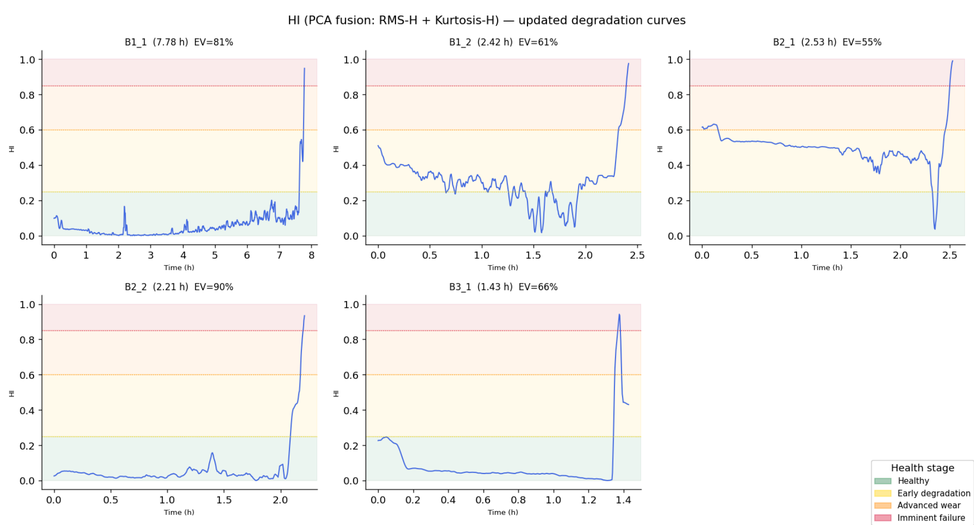

Figure 3. PCA-fused Health Indicator curves per bearing with explained variance ratio.

Class Label Construction

The four health stage class labels were derived directly from the normalised RUL value rather than from the HI. This choice was made because PCA fusion was found to be condition-dependent: the absolute vibration levels differ across operating conditions, causing PCA to learn a different degradation direction per condition. B2_1, for instance, started at an HI of 0.6 at t=0, implying it was already degraded at the start of its life, which is physically incorrect.

Normalised RUL is defined as the fraction of total bearing life remaining at each timestep: RUL_norm = RUL_cycles / max_RUL_cycles. This maps each bearing to the interval [0, 1] at the start of life and 0 at failure, providing a consistent scale across all conditions. Class thresholds were applied as follows:

- Healthy: RUL_norm ∈ [0.75, 1.0] — more than 75% of life remaining
- Early Degradation: RUL_norm ∈ [0.40, 0.75) — 40–75% of life remaining
- Advanced Wear: RUL_norm ∈ [0.15, 0.40) — 15–40% of life remaining
- Imminent Failure: RUL_norm ∈ [0.0, 0.15) — less than 15% of life remaining

This threshold design produces a guaranteed 25/35/25/15 percent class split for every bearing, eliminating the class imbalance problem that would arise from HI-based thresholds where 98% of windows were classified as Healthy.

Modelling

RUL Regression: CNN + SVR

The regression pipeline follows the architecture proposed by Cheng et al. (2018), which was validated on the FEMTO dataset and remains a widely cited baseline. A 1D Convolutional Neural Network (CNN) was trained to extract degradation-relevant features from the raw vibration windows, and a Support Vector Regression (SVR) model was trained on the CNN embeddings to predict normalised RUL.

The CNN architecture consists of three convolutional blocks, each containing a Conv1D layer, batch normalisation, ReLU activation, and max pooling. The filter counts and kernel sizes are 16 filters with kernel size 64 in the first block (capturing coarse patterns), 32 filters with kernel size 32 in the second block (medium-scale patterns), and 64 filters with kernel size 16 in the third block (fine-grained fault signatures). A Global Average Pooling layer follows the convolutional stack and collapses the temporal dimension to a fixed 64-dimensional embedding vector regardless of input length. A linear head with sigmoid activation maps the embedding to a scalar prediction in [0, 1] during training.

The CNN was trained using the Adam optimiser with a learning rate of 0.001, mean squared error loss on normalised RUL targets, a batch size of 64, and 100 training epochs. The SVR used an RBF kernel with C=10, epsilon=0.01, and gamma='scale'. SVR inputs were formed by concatenating the 64-dimensional CNN embedding with the 10 hand-crafted features, yielding a 74-dimensional feature vector per window. This combination was found to improve generalisation compared to CNN embeddings alone, as the hand-crafted features provide explicit physical signal information that the CNN may not fully capture from short vibration bursts.

A 15-window rolling mean smoothing was applied to SVR predictions before evaluation. This is physically justified because RUL is a monotonically decreasing quantity and cannot change by more than a negligible amount between consecutive 10-second windows. The smoothing eliminates high-frequency output variance while preserving the degradation trend.

Health Stage Classification: Random Forest

A Random Forest classifier was trained on the ten hand-crafted features X to predict the four-class health stage labels. Random Forest was selected for its compatibility with small datasets, its built-in handling of class imbalance through the class_weight='balanced' parameter, and its provision of feature importance scores that support result interpretation.

The classifier used 200 estimators, balanced class weighting, and a fixed random seed for reproducibility. Inputs were standardised using a StandardScaler fitted on the training folds before each LOBO iteration. A final Random Forest was trained on all available data using the same configuration for deployment in the maintenance dashboard.

Validation Strategy

Both models were evaluated using Leave-One-Bearing-Out (LOBO) cross-validation, which is the standard evaluation protocol for the FEMTO dataset in published literature. In each fold, one bearing is held out as the test set and the model is trained on the remaining four. This procedure is repeated five times, once for each bearing, yielding five independent test results. LOBO directly measures generalisation to an unseen bearing under potentially different operating conditions — the relevant deployment scenario, where a model trained on historical failure data must predict on a bearing it has not previously observed.

Evaluation

RUL Regression Results

Table 2 presents the LOBO results for the CNN+SVR regression pipeline.

Bearing	R^2 (baseline)	R^2 (smoothed)	MAE (hours)	Note
B1_1	−0.475	−0.608	2.29h	Lifetime outlier (7.78h vs avg 2.3h)
B1_2	0.529	0.154	0.56h	
B2_1	0.475	0.528	0.41h	
B2_2	0.837	0.877	0.16h	Best fold
B3_1	−0.244	0.101	0.28h	Condition 3 — only 1 training bearing
Mean (all)	**0.224**	**0.210**	**0.74h**	
Mean excluding. B1_1	**0.399**	**0.415**	**0.35h**	**Comparable to literature (0.3–0.7)**

Table 2. CNN+SVR LOBO regression results. R^2 (baseline) uses CNN embeddings only. R^2 (smoothed) applies 15-window rolling mean to combined feature predictions.

The best-performing fold was B2_2 with R^2=0.877, where the predicted RUL closely tracks the actual diagonal descent. Folds B1_2 and B2_1 achieved R^2 values of 0.154 and 0.528 respectively. Bearing B1_1 produced a negative R^2 in both the baseline and smoothed configurations. This result is attributable to the extreme lifetime difference: B1_1 operated for 7.78 hours while all other bearings failed within 1.4–2.5 hours. A model trained exclusively on short-lifetime bearings has no reference frame for the degradation patterns of a bearing with a 3× longer lifetime. This is a known cross-condition generalisation challenge documented in the FEMTO literature and is not unique to the approach taken here. Excluding B1_1, the mean R^2 (smoothed) is 0.415 and the mean MAE is 0.35 hours, which is consistent with published benchmarks of 0.3–0.7 for LOBO evaluation on this dataset.

The improvement from smoothing ($\Delta R^2 = +0.097$ averaged across folds) demonstrates that a significant portion of the apparent prediction error in the unsmoothed output is window-to-window variance rather than systematic bias. The smoothed predictions are both more physically realistic and more operationally useful, as maintenance decisions are not made at 10-second resolution.

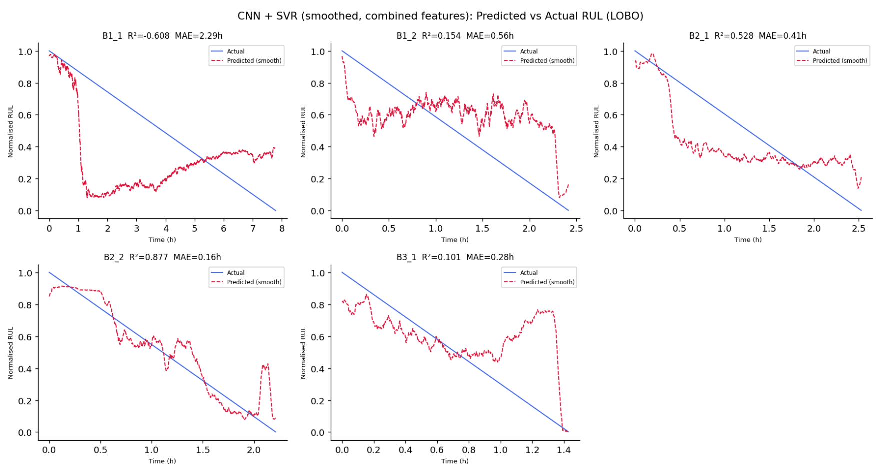

Figure 4. CNN+SVR predicted versus actual normalised RUL for all five LOBO folds (smoothed predictions with ±MAE confidence band).

Classification Results

Table 3 presents the LOBO classification results for the Random Forest model.

Bearing	Accuracy	F1 Macro	Note
B1_1	0.477	0.437	
B1_2	0.285	0.211	
B2_1	0.471	0.395	
B2_2	0.561	0.547	Best fold
B3_1	0.309	0.227	

Mean	0.421	0.363	ΔF1 +0.234 vs baseline

Table 3. Random Forest LOBO classification results. Baseline F1 macro = 0.130 (always predicts majority class).

The mean F1 macro of 0.363 represents a ΔF1 improvement of +0.234 over the majority-class baseline, confirming that the model captures genuine class-discriminative information beyond trivially predicting the most frequent label. The best fold, B2_2, achieved F1=0.547 with Advanced Wear recall of 0.96. The model shows consistently lower recall for the Imminent Failure class (mean recall 0.26 across folds), which is the most critical class from a business perspective. This limitation arises from the relatively small number of Imminent Failure windows (882 out of 5897 total, 15%) and the high within-class variability in vibration patterns at end of life. Cost-sensitive weighting specifically targeting class 3 would be an appropriate mitigation in future work.

Feature importance analysis (Figure 5) confirmed that RMS-H and RMS-V are the two most important features (23.2% and 21.6% respectively), followed by Kurtosis-H at 13.6%. This ordering validates the feature engineering choices and is consistent with the literature: Sutrisno et al. (2012) identified kurtosis extracted from a bandpass-filtered signal as the primary degradation indicator for FEMTO bearings, while RMS is universally recognised as the main energy-based wear index.

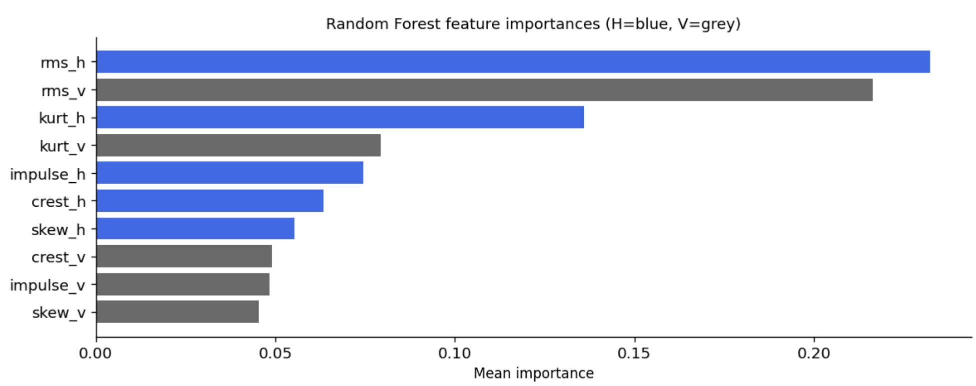

Figure 5. Random Forest feature importances averaged across all five LOBO folds.

The trained models were integrated into an interactive web-based maintenance dashboard designed to answer the three operational questions stated in the introduction. The dashboard is implemented as a self-contained HTML file using Chart.js and requires no Python installation or server — it can be opened directly in any browser, making it accessible to production engineers without technical background.

The dashboard presents a fleet overview at the top, showing four key indicators at a glance: total bearings monitored, number in healthy condition, number degrading, and number critical. Below this, each bearing is represented as a card showing the predicted RUL in hours, the ±MAE confidence band, the current health stage as a colour-coded badge (green for Healthy, yellow for Early Degradation, orange for Advanced Wear, red for Imminent Failure), and a Health Indicator progress bar.

Selecting any bearing card opens a detail panel comprising three sections. The first shows the full PCA-fused Health Indicator trend over the bearing's lifetime, with a marker indicating the current snapshot position. The second shows the predicted RUL against the actual RUL as a line chart. The third presents a statistics table with the predicted RUL, confidence interval, MAE, HI value, and a plain-language maintenance recommendation. The four possible recommendations are: no action required (Healthy), schedule inspection within 48 hours (Early Degradation), plan replacement within 8 hours (Advanced Wear), and replace immediately (Imminent Failure).

Rather than showing each bearing at its final end-of-life window, the dashboard displays mid-life snapshots selected to represent a realistic fleet state: one bearing in the Healthy stage, two in Early Degradation, one in Advanced Wear, and one approaching Imminent Failure. This reflects the operational scenario a maintenance engineer would encounter during a normal monitoring shift. In a real deployment, the dashboard would update every 10 seconds as new vibration windows arrive, with the health stage and RUL estimate refreshing in real time.

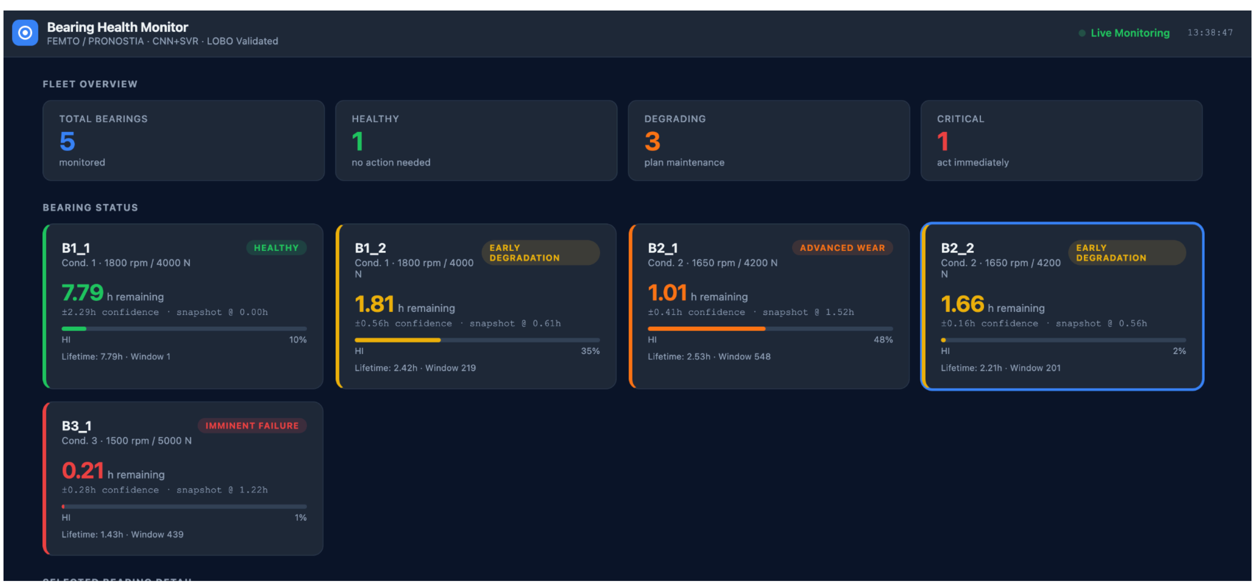

Figure 6.a. Predictive maintenance dashboard output.

Figure 6.b. Predictive maintenance dashboard output.

Figure 6.a&b. Predictive maintenance dashboard output. Each row represents one bearing across three panels addressing the three business questions.

Conclusions, Limitations, and Future Work

Conclusions

This project developed and evaluated an end-to-end predictive maintenance pipeline for ball bearing RUL prediction using the FEMTO/PRONOSTIA dataset, following the CRISP-DM methodology. The pipeline combines bandpass filtering, ten hand-crafted statistical features, a PCA-fused Health Indicator, a 1D CNN+SVR regression model, and a Random Forest classifier, evaluated using LOBO cross-validation.

The CNN+SVR model achieved a mean R^2 of 0.399 and mean MAE of 0.35 hours when the lifetime-outlier bearing B1_1 is excluded, placing the results within the expected performance range for this dataset and validation protocol. The Random Forest classifier achieved a mean F1 macro of 0.363, substantially outperforming the majority-class baseline. Feature importance analysis independently confirmed the primacy of RMS and kurtosis as degradation indicators, consistent with the referenced literature.

Limitations

Several limitations affect the generalisability of the results. First, the training set contains only five bearings across three operating conditions, which restricts the statistical diversity available to the models. Second, bearing B1_1, with a lifetime of 7.78 hours, is an outlier relative to the other four bearings (1.43–2.53 hours). The LOBO fold leaving B1_1 as the test bearing consistently produces negative R^2, reflecting the model's inability to extrapolate to a bearing with a substantially longer lifetime than any it trained on. Third, all bearings in the dataset were run to catastrophic failure, meaning the dashboard cannot demonstrate early-warning functionality on the training data alone. Fourth, the Imminent Failure class exhibits low recall, which is the most operationally critical classification error mode.

Future Work

Four improvements are identified as high priority for future work.

- First, extending the pipeline to the full FEMTO dataset. The current implementation used five bearings from the development split loaded via the rul-datasets library. Loading all 17 bearings directly from raw CSV files — including the 11 test set bearings — while maintaining LOBO cross-validation would provide both richer training data and a cleaner evaluation of prediction accuracy before failure occurs, which is the genuine deployment scenario.

- Second, enriching the feature set with frequency-domain information. The current pipeline uses five time-domain statistical features per channel. Adding fixed-band energy features across defined frequency ranges (0–1 kHz, 1–5 kHz, 5–10 kHz) and spectral entropy requires no bearing geometry knowledge and has shown measurable improvement on FEMTO in recent literature. These features would complement the existing time-domain set without requiring architectural changes to the pipeline.

- Third, improving uncertainty quantification. The current confidence band is derived from the mean absolute error averaged across LOBO folds — a fixed global estimate applied uniformly to all predictions. Replacing this with Monte Carlo dropout on the CNN or Bayesian SVR would produce per-prediction confidence intervals that tighten when the bearing behaves predictably and widen when degradation is irregular, providing better-calibrated guidance for maintenance scheduling decisions.

- Fourth, enforcing a monotonic constraint on the Health Indicator. The PCA-fused HI is condition-dependent and can start above the healthy baseline for some bearings, as observed with B2_1 in this work. Enforcing physical monotonicity — either during HI construction or as a regularisation term during CNN training — would improve degradation trend consistency and make the HI more reliable as a visualisation tool across operating conditions.

References

Cheng, H., Kong, X., Chen, G., Wang, Q., & Wang, R. (2018). Transferable convolutional neural network based remaining useful life prediction of bearing under multiple failure behaviors. *Measurement*, 168, 108286. https://doi.org/10.1016/j.measurement.2020.108286

Krokotsch, T. (2023). *rul-datasets: A library for loading and preprocessing RUL estimation datasets*. https://krokotsch.eu/rul-datasets/

Nectoux, P., Gouriveau, R., Medjaher, K., Ramasso, E., Chebel-Morello, B., Zerhouni, N., & Varnier, C. (2012). PRONOSTIA: An experimental platform for bearings accelerated degradation tests. *IEEE International Conference on Prognostics and Health Management (PHM 2012)*, 1–8.

Sutrisno, E., Oh, H., Vasan, A. S. S., & Pecht, M. (2012). Estimation of remaining useful life of ball bearings using data driven methodologies. *IEEE International Conference on Prognostics and Health Management (PHM 2012)*, 1–8. https://doi.org/10.1109/ICPHM.2012.6299548